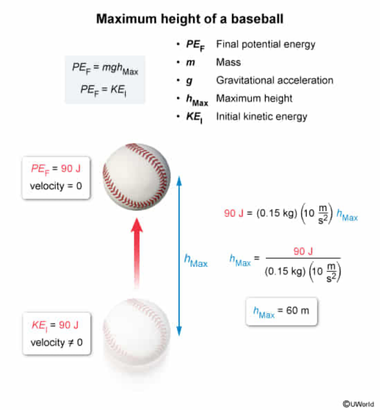
What is the maximum possible height of a 0.15 kg baseball if it has 90 J of kinetic energy after being hit by a baseball bat?

- A. 130 m [7%]
- B. 90 m [8%]
- C. 60 m [78%]
- D. 9.0 m [6%]

Correct
79% answered correctly

Explanation:



The total mechanical energy E of an object in motion equals the sum of its kinetic energy KE and the system's potential energy PE:

$$E = \mathrm{KE} + \mathrm{PE}$$

Energy cannot be created or destroyed, but it can be transformed from one form into another. In the case of an object in motion, energy can be transformed from KE to PE and vice versa, but the total energy must be conserved (ie, remain constant):

$$E = \mathrm{KE} + \mathrm{PE} = \text{constant}$$

Gravitational potential energy equals the product of an object's mass m, gravitational acceleration g, and its height h:

$$\mathrm{PE} = \mathrm{mgh}$$

In this question, the baseball has an initial kinetic energy KE_I that is completely converted to PE when the ball moves straight upward and reaches its maximum possible height h_{Max}. Conservation of energy implies that KE_I equals the final potential energy PE_F:

$$KE_I = PE_F = \mathrm{mgh}_{\mathrm{Max}}$$

The baseball has KE_I of 90 J and m of 0.15 kg. Substituting these values into the above equation and solving for h_{Max} yields:

$$h_{\mathrm{Max}} = \frac{KE_I}{\mathrm{mg}} = \frac{90 \text{ J}}{(0.15 \text{ kg})\left(10 \frac{\text{m}}{\text{s}^2}\right)} = 60 \text{ m}$$

Therefore, the baseball will reach a h_{Max} of 60 m.

(Choice A) 130 m is obtained from the product of the kinetic energy, mass of the ball, and gravitational acceleration.

(Choice B) 90 m results from using 0.10 kg for the mass of the ball instead of 0.15 kg.

(Choice D) 9.0 m is calculated from the ratio of the kinetic energy of the ball and gravitational acceleration.

Educational objective:
Conservation of energy states that the total mechanical energy of an object, which equals the sum of potential energy and kinetic energy, must remain constant. The maximum height an object moving straight upward reaches can be calculated from its initial kinetic energy.

Time spent:0 sec
Subject:
Physics

QID:404153
System:
Mechanics and Energy